

RADAR-ACTUATED LAWNMOWING PROTECTIVE HOUSING

Lawn Mower Safety Device

Advisors: Jason Forman, Michael Momot

Team: Richard Townsley, Jackson Berry, Jimmy Sejas, Vincent Hu,
Averell Stith, David Cuyuch, Mia Bonutti, Cole Smith

Date: April 28, 2026

INTRODUCTION

Across the United States, millions of households rely on lawnmowers for home maintenance. While usually considered a safe device, the high-speed rotating blades can cause severe trauma, including amputations and death, primarily with subjects under the age of 18 (Jennissen, et al., 2023). Despite safety improvements over the past few decades, the number of lawnmower related injuries has remained consistent with an average of seventy-five fatalities per year for children (Jennissen, et al., 2023). A disproportionate number of children are affected by lawnmower accidents with an estimated 10,000 children affected annually with a peak in injury rates at ages 4 and 15 (Jennissen, et al., 2023). To help contextualize these alarming numbers, over 130,000 children visit the ER for bicycle related injuries while only around one hundred are fatally injured each year (Safekids, 2020).

Given this background, the Lawnmower Safety Design Group has chosen to address the issue of the dangers posed by lawnmower blades to bystanders. Our primary concern is reducing injuries resulting from ride-over accidents where a bystander ends up underneath the mower and is injured by the blades. While important to address the youth, the team deemed it was just as important to include all bystanders to best mitigate risk of injury. Given this background, the team has concluded that a viable solution must be non-discriminative and fully preventative in nature to protect all bystanders from injury from lawnmower blades.

RESEARCH

The lawnmower industry has implemented several innovations to prevent injuries from ride-over events. Some common features include various operator presence control (OPC) systems. These are systems that stop the blades if the operator leaves the lawnmower seat or removes their hands from the controls. Some mowers feature blade brakes, but this mechanism

has not been widely adopted for riding lawn mowers. Many modern riding lawn mowers enable the user to stop or activate the blades via a lever while allowing the engine to power the mower. This increases safety by allowing the user to only engage the blades when necessary. Despite these safety innovations, lawn mower injuries have not decreased for several decades in the United States (Khansa et al., 2021). Previous patented solutions have involved proximity sensors along with programmed instructions to vary blade behavior or apply brakes to the wheels to prevent continued movement. (Bernini, 2009). Despite the patent being filed in 2008, it has not seen widespread commercial use indicating that it was likely unsuitable for commercial production.

IDEATION

In the ideation stage, the team decided to individually produce ten ideas each to tackle the issue of lawnmower blade injury prevention. In this stage, each member was encouraged to produce unique and creative ideas. While this process brought up farfetched ideas, like having a net that is deployed upon a person entering a defined radius around the lawnmower, the process also brought up extremely well-thought-out ideas. Some of the ideas are as follows: a high-powered braking mechanism, a spring-loaded retractable blade assembly, as well as reciprocating blades.

The team also collectively produced ten key specifications that allowed us to score each idea with an objective lens. These ten specifications served as a tool to help us grade our ideas individually, and then present our ideas accordingly based on what each individual thought was impactful. Ultimately, the use of individual discretion in the ideation process as well as using our specifications accordingly allowed the team to concretely move onto selection and screening. The ten specifications are as follows:

1. Design entirely prevents 100% of user contact with the cutting instrument of the lawn mower to avoid injury in optimal conditions.
2. Must not destroy lawn mower when activated and must be reusable and resettable.
3. Must be activated from start-up to shut-down.
4. Does not impact operation, maneuverability, or effectiveness of the lawn mower.
5. Does not require extra input from the users such as new buttons, options, features, etc.
6. Electrically and mechanically robust enough for typical industry standard lawn mowing conditions, including wet grass. Must last a typical lawn mower life cycle of 5-15 years.
7. Must not make maintenance considerably more difficult, complex, or time consuming compared to original maintenance.
8. Does not cost more than 20% of the average riding mower price of \$3,940.
9. The average person should be able to maintain lawn mower system using simple manual tools and readily available parts.
10. Must be biased towards false positives but must not trigger incorrectly more than two times per hour.

SELECTION AND SCREENING

In the selection and screening phase, each person pitched three of their ideas from the ideation phase, in respect to what was most technically feasible as well as realistic. From this informal pitch, the team narrowed the workable solutions to ten ideas. Additionally, the team elected to go through a preliminary scoring system that included the current state of a commercial ride-on lawn mower as a control. This scoring system allowed us to eliminate some outliers and is shown below in *Figure 1*.

Category	Concepts									
	Control (Riding Mower)	Hydraulic Lift	High Powered Braking	Deployable Guards	Spring-Loaded Lift	Flowbee Design	Safety Blades	Helical Blades	Reciprocating Blades	Grazing System
Technical Feasibility	0	-1	-1	-1	-1	-1	-1	-1	-1	-1
Ease of Manufacturing	0	-1	-1	0	-1	-1	-1	0	0	-1
Cost	0	-1	-1	0	-1	-1	0	0	0	-1
Cutting Experience	0	0	0	0	0	-1	-1	1	-1	0
Safety	0	1	1	1	1	1	1	1	1	1
Durability	0	-1	-1	1	0	-1	-1	0	0	0
Ease of Maintenance	0	-1	0	0	-1	-1	-1	-1	-1	-1
Total	0	-4	-3	1	-3	-5	-4	0	-2	-3
Continue?		No	Yes	Yes	Yes	No	No	Yes	Yes	Yes

Figure 1. Preliminary Scoring System

In the above image, the team decided on seven categories that were deemed important in delivering both an impactful and reliable product. These categories can be seen on the left-hand side under the “Category” column. They include technical feasibility, cost, ease of maintenance, and other relevant metrics. With a tick of –1 for a no, a tick of 0 for neutral, and a tick of 1 for yes, the team decided to eliminate design ideas that consisted of scores of –3 or more. The remaining three ideas — the hydraulic lift, Flowbee design, and safety blades — were ruled out during final design consideration.

In the final design consideration stage, the team created a more robust scoring system, one which allowed for more visibility on certain ideals that were considered more pertinent to the final design. This scoring system placed a high emphasis on individual weighting of certain categories. Furthermore, instead of rating with a no, neutral, or yes system as reflected by the tick scoring method, the team chose to rate each category on a scale of 1-5 to gain more visibility as well as a better understanding of each design's advantages, including ranking the control. That system is shown in *Figure 2*.

Category	Weight	Riding Mower		Spring-Loaded Lift		Grazing		Helical		Trapdoor		Brake		Reciprocating Blade	
		Rating	Score	Rating	Score	Rating	Score	Rating	Score	Rating	Score	Rating	Score	Rating	Score
Technical Feasibility	20%	5	1	4	0.8	2	0.4	3	0.6	4	0.8	3	0.6	4	0.8
Ease of Manufacturing	5%	4	0.2	3	0.15	2	0.1	4	0.2	4	0.2	2	0.1	3	0.15
Cost	10%	3	0.3	3	0.3	2	0.2	3	0.3	3	0.3	2	0.2	3	0.3
Cutting Experience	20%	5	1	4	0.8	2	0.4	4	0.8	3	0.6	4	0.8	3	0.6
Safety	30%	1	0.3	3	0.9	4	1.2	3	0.9	4	1.2	4	1.2	4	1.2
Durability	10%	3	0.3	3	0.3	3	0.3	3	0.3	3	0.3	2	0.2	3	0.3
Ease of Maintenance	5%	4	0.2	3	0.15	2	0.1	3	0.15	3	0.15	2	0.1	3	0.15
Total	100%		3.3		3.4		2.7		3.25		3.55		3.2		3.5

5 is best for each category

Figure 2. Final Design Scoring System

Ultimately, after scoring, the team picked the “Trapdoor” idea as the most feasible design as it ranked amongst the highest in both technical feasibility and ease of manufacturing. Additionally, this design was scored as being the most impactful and the largest improvement over the control.

TECHNICAL ASPECTS, PROTOTYPES, FINAL SPECIFICATIONS

The core mechanisms can be categorized into five systems: a wheeled base, track and guard, release, retraction, and a sensor. We constructed a 5-inch tall, wheeled box with 30 -inch sides to represent the mower and to serve as a mounting surface for other system components. The rail-guard system consists of two ball bearing carriages accompanying 130 mm rails. The carriages are attached to a 30-inch x 5-inch x 0.125-inch polycarbonate board which functions as a guard to prevent contact with the mower blades. 3D printed stops underneath the rails prevent the carriages from traveling too far and detaching from the track. Initially, the guard is held off the ground by an electronic rotary release. When the millimeter wave (mmWave) sensor detects an object, the electronic rotary release activates and allows the guard to travel down the rails until it contacts the ground. The sensor is installed within a 3D-printed sensor mount that ensures the sensor has an unobstructed view in front of the mower. After ensuring the mower is clear of obstructions, the guard can then be retracted to its original position with a stepper motor and ball

screw assembly which pulls the guard up until the latch engages and then moves out of the way to resume operation.

WHEELED BASE

To emulate the geometry of a lawn mower blade housing and the movement of a lawn mower, we designed a 30"×30"×5" box, with 2x4s for inner reinforcement and four caster wheels at the bottom. Fastened together using wood screws, the sides and top of the base were initially made of 1/8" plywood. This material was not structurally sound and could cause alignment and bowing issues during assembly. We opted to completely remake the box with 3/4" MDF wood to ensure structural integrity.

RAIL-GUARD SYSTEM

Our initial plan was to make our guard out of aluminum, but we eventually opted to use polycarbonate. This was for demonstration purposes, as the material's transparency allows viewers to see the rail system behind the guard. The polycarbonate was cut to 30"×5" to match the dimensions of the front of our base. We chose a thickness of 1/8" for our demonstration as it was cheap and a common size.

The team first chose to calculate the force produced by an adult kick. The team defined the mass of an adult foot as 0.635kg, the mass of the two-body system as 2.5kg, assumed an average velocity of 9 m/s per adult kick, and assumed a contact time of 10ms per the duration of average contact from a soccer ball. After this the team used the impulse force calculation, $F_{avg} = \frac{m\Delta v}{\Delta t}$, which resulted in 571 N of average force. We further applied a dynamic load factor of 2.0, giving a peak force of 1143N. The team then calculated the deflection of the guard, shown below in *Figure 3*. The max deflection was 3.8mm for a 1/4" aluminum guard, which is the best material for the environment as it resists rust and is fairly lightweight. While the FEA did show stress

concentrations higher than yield strength at the corners, we determined this was a limitation of the static assumption given these stress concentration would not exist to that degree in real life as the guard would not be held perfectly rigidly, and so some of this stress would be taken up by moment within the bolts and vibration. While our polycarbonate example is not strong enough to resist a kick, it was quite strong enough at full speed drop tests against a concrete floor, which is harsher than the typical use case on grass, and thus we feel confident that an aluminum guard would be more than sufficient for typical use.

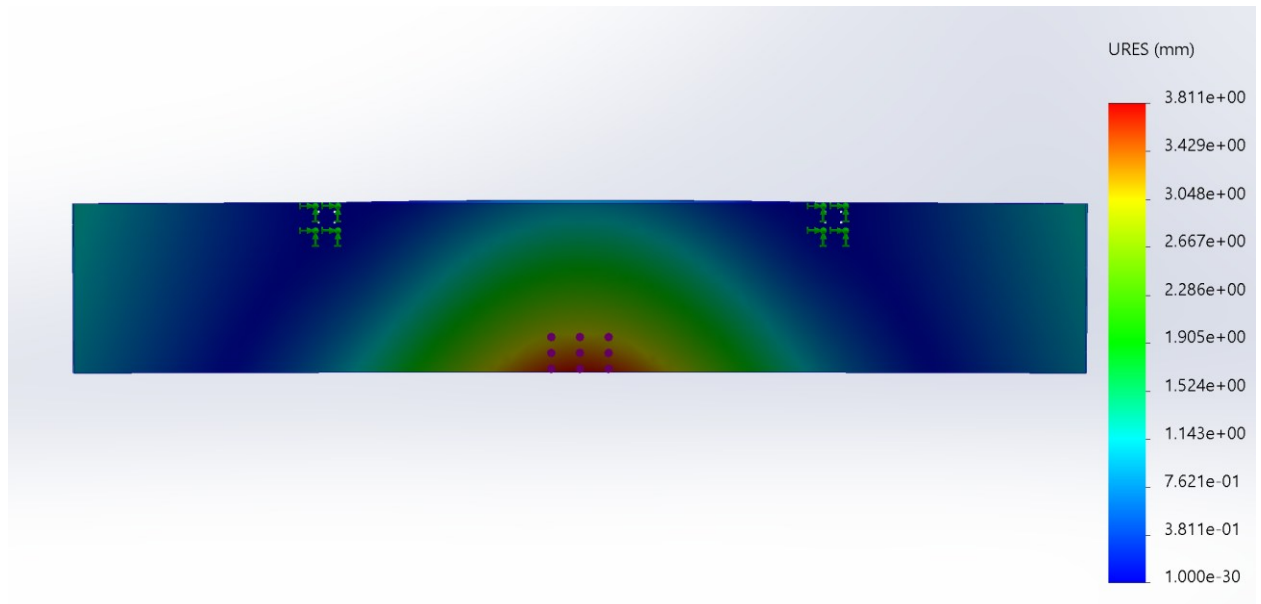


Figure 3. Deflection of guard under dynamic impact

To control the vertical movement of the guard, we opted to use a linear rail system. We chose 7mm rails and corresponding ball-bearing carriages because they were the cheapest options we could find and were designed for low friction. The small size of the rails and carriages did lead to issues with alignment. Both the rails and carriages used M2 screws, and the alignment of these parts was critical because if the rails were not placed perfectly vertically and if the carriages were not perfectly aligned with the rails, it could prevent the guard from falling at

its maximum speed. We opted to have Kemal Goturk, the head of the UVA Machine Shop, machine the holes in our guard as well as in front of our base to ensure they were all aligned correctly. Once the M2 holes were correctly drilled we attached the carriages, placed symmetrically and 15” apart, to the guard using M2 screws, washers, and nuts. We eventually placed 3-D printed spacers made of PETG between the carriages and the guard to create space for the retraction and spring systems.

The final piece of the rail-guard system is the stops. In a typical lawn mowing situation, the guard will be stopped by the guard hitting the ground. However, if the lawn mower is not mowing on a flat surface and there is a large drop in elevation at the front of the lawn mower, there could be a risk of the guard sliding off the rails entirely. Stops were added to the bottom of each of the rails to prevent this. Initially, the stops were 1/8” 3-D printed L-brackets, placed flush to the bottom of the base. Eventually, the stops were changed to be fastened to front of the base at the bottom portion of the rails to be compatible with the coupler of the retraction system, as the coupler needs to be below the stops during release. The release and retraction system will be discussed in greater detail later in this section. The stops were attached to the front of the base using 1 1/4” screws, nuts, and washers, and were 3-D printed using PETG. All final dimensions and assembly features of the rail-guard system can be seen in *Appendix A, Fig. A-1*.

RELEASE SYSTEM

The release system consists of three primary components: the latch, the hook, and the spring. For the latch mechanism, an electronic rotary latch was selected due to its ability to resist vibrations compared to linear latches. This is an important consideration since the continuous vibration present in typical lawnmower operating environments caused by uneven terrain and fast spinning blades. The initial rotary latch considered for the design was gear-driven, which

resulted in it being too slow for our purposes and introduced unnecessary cost. To ensure proper operation, the latch must release the guard quickly enough for full deployment fast enough to block a human limb. We decided on a 1m detection radius with an individual traveling at 3.5m/s, representing a person walking quickly toward a mower also being pushed quickly. Since $t = d/v$, this gives a total response time of 285ms.

The rotary latch we decided on is solenoid driven from ATOPLEE, which can be activated electronically and releases nearly instantly. Along with this, this model is cheap, ~\$16 per unit and able to withstand the 23.22N of net force applied from the guard and springs pushing down on it. To attach the latch to the wooden box, a 3D printed L-bracket was made of PETG to accommodate the retraction system alongside it. The latch is oriented vertically with the wires and has an upward-facing mechanical unlocking lever in the event of electrical failure. The latch is connected via crimped wires to the microcontroller board and unlocked when the mmWave sensor detects a person within 1m. The 3D connection piece can be seen in *Appendix A, Fig. A-7*.

The hook component is connected to the guard using an aluminum U-channel which allows us to place the hook on the top of the guard and closer or further away from the blade housing box we made. The U-channel rests on top of the guard with two bolts going through the flanges with nuts on either side of the outside flange to secure the guard in the desired location. The guard ended up being manufactured out of polycarbonate instead of aluminum and the distance a hole in plastic materials should be at least a distance of 1.5x the diameter of the bolt away from the edge of the material (United States Plastic Corp, 2023). However, due to the small width of the U-channel flanges, the holes could not be drilled 1.5x the diameter of the hole

length away; to add strength, a 1/16in piece of aluminum was used to reduce stress and prevent shearing of the polycarbonate guard. The U-channel can be seen in *Appendix A, Fig. A- 8*.

For the spring component, compression springs were selected since they can reliably store the required potential energy and deliver the necessary downward force to deploy the guard within the required time. The initial design calculations specified springs with a stiffness of 3852N, a maximum compression of 0.0396m (1.561”), and a minimum compressed length of 0.1016m (4”) to ensure an extension of 2” during guard deployment. With one spring mounted on each carriage for two total springs, the combined maximum force required was calculated to be 477N. The detailed calculations supporting the initial spring selection are provided below in *Figure 4*.

Spring Calculations Old

Known/Given

- Spring rate = $k = 22 \frac{\text{lb}}{\text{in}} \cdot \frac{1\text{m}}{0.0254\text{m}} \cdot \frac{4.448\text{N}}{1\text{lb}} = 3852.598 \frac{\text{N}}{\text{m}}$
- Length of Spring compressed = $L_c = 1.561\text{in} \cdot \frac{0.0254\text{m}}{1\text{in}} = 0.0396\text{m}$
- Normal length of Spring = $L = 4\text{in} \cdot \frac{0.0254\text{m}}{1\text{in}} = 0.1016\text{m}$
- Vertical distance guard covers = $d = 2\text{in} \cdot \frac{1\text{m}}{39.37\text{in}} = 0.0508\text{m}$

Calculations

Max Spring Force = $F_{\text{max}} = k (L - L_c) = 3852.598 \frac{\text{N}}{\text{m}} (0.1016 - 0.0396\text{m}) = 238.34\text{N}$

Min Spring Force = $F_{\text{min}} = k [L - (L_c + d)] = 3852.598 \frac{\text{N}}{\text{m}} [0.1016 - (0.0396 + 0.0508\text{m})] = 85.92\text{N}$

total force for 2 springs

Figure 4. Calculations of Initial Spring Requirements.

After receiving the initial springs, we determined that the springs were excessively stiff and would exceed the torque capacity of the retraction motor when resetting the guard and cause significant deflection of the guard. A review of our earlier analysis showed that our design assumptions were overly conservative, resulting in an unnecessarily high spring force requirement. The calculations were subsequently revised using updated loading conditions and using a more accurate calculation method treating the spring as a harmonic oscillator, yielding a

new allowable maximum spring force of 55.7N split between two springs, and an estimated drop time of 31.7ms. The resulting spring specifications based on these updated calculations are presented below in *Figure 5*, which shows the spring force calculation, in *Figure 6*, which shows our calculations of drop time, and *Figure 7*, which shows a graph of guard height vs time and the intersection point where the guard will come to a stop.

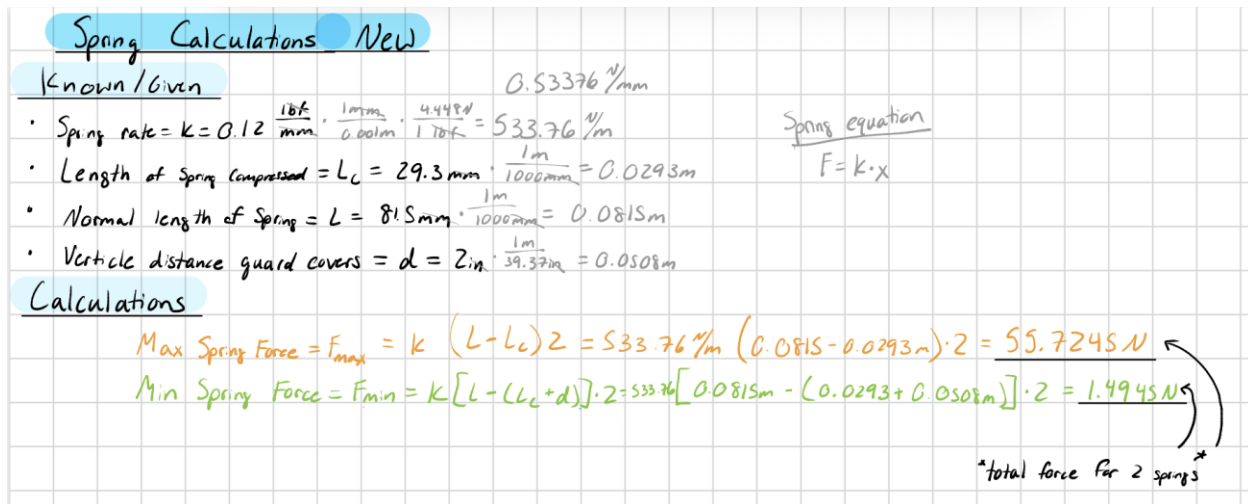


Figure 5. Calculation of Maximum and Minimum Spring Forces for New Spring.

$$y(t) = y_c(t) + y_p(t) = C \sin(\sqrt{k/m} t) + D \cos(\sqrt{k/m} t) + \frac{mg - F}{k}$$

$$y(0) = y_{\text{min}} = C \sin(0) + D \cos(0) + \frac{mg - F}{k}$$

$$y_{\text{min}} = D + \frac{mg - F}{k} \rightarrow D = y_{\text{min}} - \frac{mg - F}{k}$$

$$y'(t) = \sqrt{\frac{k}{m}} \cdot C \cos(\sqrt{k/m} t) - D \sqrt{\frac{k}{m}} \sin(\sqrt{k/m} t) + \frac{(mg - F)}{k}$$

$$y'(0) = 0 = \sqrt{k/m} \cdot C \rightarrow C = 0$$

$$y(t) = \left(y_{\text{min}} - \frac{mg - F}{k} \right) \cos(\sqrt{k/m} t) + \frac{mg - F}{k}$$

$mg_{\text{guard}} = \rho V = 0.762 \text{ m} \cdot 0.127 \text{ m} \cdot 0.00318 \text{ m} \cdot 1200 \frac{\text{kg}}{\text{m}^3} = 0.3671 \text{ kg}$
 $mg_{\text{spring}} = \rho V = \rho \cdot A \cdot L = \rho \cdot \pi r^2 \cdot \pi D N t, D = \frac{3}{8} D_o + D_c, N t = \frac{G \cdot d^4}{8 D_o^3 k}$
 $= 7850 \text{ kg/m}^3 \cdot \pi (0.001 \text{ m})^2 \cdot (0.01 \text{ m}) \cdot 79 \cdot 10^9 \text{ Pa} \cdot (0.001 \text{ m})^4 = 1.14 \text{ g each so}$
 $840.01 \text{ m}^3 \cdot 533 \text{ N/m}$ Spring mass is negligible

$y_{\text{max}} = -2'' = -0.05 \text{ m}$
 $k = 0.12 \text{ lb/f/mm} = 533 \text{ N/m} \cdot 2 \text{ springs} = 1066 \text{ N/m}$
 $F = 10 \text{ N}$ $g = 9.8 \text{ m/s}^2$
 Find t when $y(t) = 0$ using desmos
 $t = 31.7 \text{ ms}$

$\sum F_y = F_g + F_s - F_{\text{fr}}$
 $mg + (k)y - F = m\ddot{y}$
 $m\ddot{y} + ky = mg - F$

* F_{fr} is from rails, independent of speed and generally a small load.

→ complementary solution $y(0) = y_{\text{min}}, y'(0) = 0$

$m\ddot{y} + ky = 0$
 $m r^2 + k r = 0, r = \pm \sqrt{k/m}$
 $y(t) = C \sin(\sqrt{k/m} t) + D \cos(\sqrt{k/m} t)$
 $r(x) = mg - F$
 $y_p(t) = A, y_p'(t) = 0, y_p''(t) = 0$
 $m\ddot{y} + ky = mg - F$
 $m(0) + k(A) = mg - F$
 $A = \frac{mg - F}{k}$

Figure 6. Spring Drop Time Calculations for New Spring.

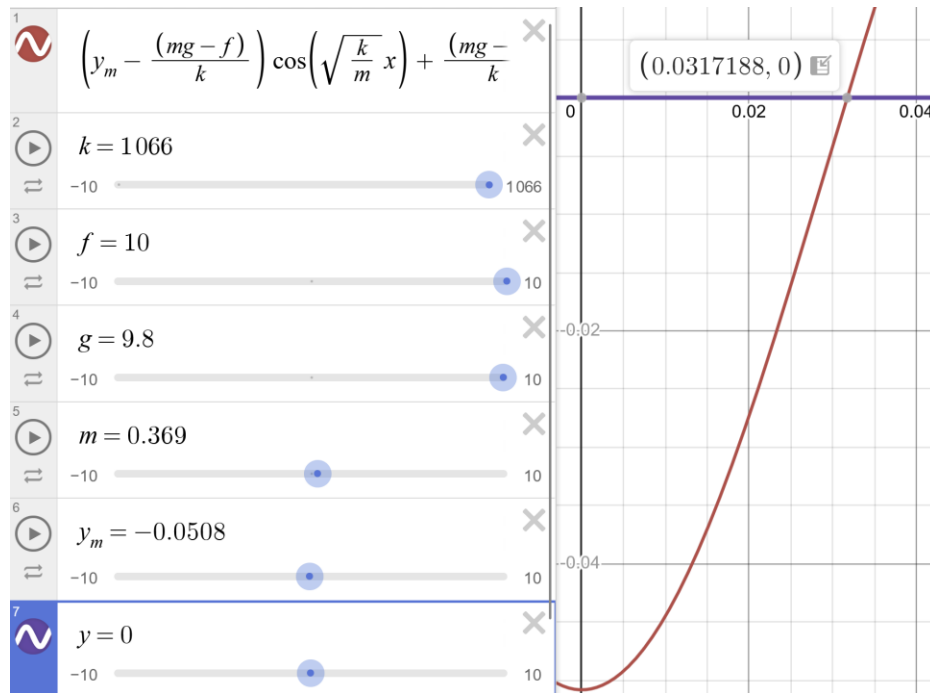


Figure 7. Graph of Guard Height vs Time

RETRACTION SYSTEM

Upon the deployment of the spring, a mechanism is required to reset the guard. Given the need to overcome spring force and our desire to make the design as convenient as possible for the user, an automatic method was devised. This mechanism consisted of a stepper motor, a ball screw, a nut, and a linkage to connect to the guard. Upon activation of a button, the motor spins the nut vertically upwards, contacting the linkage, which pulls the guard up until the latch clicks into place. A limit switch was glued to the top of the latch such that the U-bolt at the top of the guard will contact the switch after it is latched. At this point, the limit switch closes a circuit which the ESP32 detects, and a signal is sent to the motor to reverse direction. The nut then is lowered out of the way of the guard, just below the stops, ensuring it will not interfere with the release mechanism.

The stepper motor is a NEMA 17 with 42N*cm of torque run at its maximum 24V, and was chosen for precise positional control, high torque at low speeds without needing gear reductions, and low cost. This specific model was selected to achieve the required torque to overcome the force of the springs, calculated below in *Figure 8*.

$$\begin{aligned}
 &\text{Force to overcome } F = F_g + F_s + F_{fr} \\
 &F_g = mg = 0.369 \text{ kg} \cdot 9.8 \text{ m/s}^2 = 3.616 \text{ N} \\
 &F_s = F_{s\max} = kx = 0.0508 \text{ m} \cdot 533 \text{ N/m} \cdot 2 = 54.15 \text{ N} \\
 &F_{fr} = \sim 10 \text{ N} \\
 &F_{\text{total}} = 67.77 \text{ N} \\
 &T_{\text{screw}} = \frac{F \cdot l}{2\pi \cdot \eta}, \quad l = 2 \text{ mm}, \quad \eta = 0.9 \\
 &T_{\text{screw}} = \frac{67.77 \text{ N} \cdot 0.002 \text{ m}}{2\pi \cdot 0.90} = 0.023 \text{ N}\cdot\text{m} \\
 &\text{At 300 RPM, NEMA 17 has } 0.350 \text{ N}\cdot\text{m torque} > T_{\text{screw}}
 \end{aligned}$$

Figure 8. Motor torque calculations

As seen in the calculations, we also checked the torque-rpm curve provided to ensure we have sufficient torque at our targeted retraction speed, as stepper motors lose torque upon higher rotational speed. The NEMA17 is controlled using a TMC2209 Adafruit motor driver breakout board, which receives commands from the ESP32. The TMC2209 was chosen as it is a popular, low-cost board with several control options, existing Arduino library support, and can supply the 24V and up to 2A required for the motor. Some of the possible control options include stall detection, current monitoring, positional feedback, and higher torque modes. While basic control is possible without serial communication, this motor driver also has a UART control mode allowing for fine-tuning of the stepper motor settings, including changing the number of microstep and torque modes. By default, the motor uses *StealthChop*, which reduces the noise produced, but also limits the torque. Through the UART interface, we disabled this, enabling *SpreadCycle* instead to achieve better torque and finer stepping. One quirk of this board is that it

uses a single line for UART, so the Tx and Rx pins of the ESP32 had to be connected through a 1kOhm resistor to send and receive serial communications. Both the *AccelStepper* and TMC2209 Arduino libraries were used to allow smooth control over the motor.

Our initially chosen springs, as mentioned in the previous section, had a lot more torque to overcome, and so an 8mm ballscrew was chosen initially to minimize friction compared to a leadscrew and reduce the load on the motor. The ballscrew came with a nut to traverse the length of the screw, and an additional 5mm to 8mm shaft adapter was used to secure it to the motor shaft, and to block the end of the shaft, held using set screws. While ideally a thrust and radial bearing would have been used at the end to manage shaft forces, we found this to be unnecessary with the weaker springs as deflection was minimal. One issue encountered with the ballscrew was back driving, where the nut had so little friction it would simply spin in place instead of moving along the screw. While higher end ballscrew nuts are pre-tensioned to avoid this issue, our cheaper nuts were not. To solve this issue, a strip of rubber was glued to the MDF on the guard, and the motor was placed such that the nut rubbed against the edge of this rubber as it traversed the shaft, providing the minimal friction required to engage without a load. This method is not ideal for a future design and should be replaced with a properly tensioned nut.

The motor and ballscrew assembly were secured with a 3D printed mount which screwed into the top wood and used smaller machine screws through the threaded holes on the motor itself. While our initial design had the motor mounted centrally, ultimately this would have made the latch hold the guard asymmetrically, causing it to tilt and release unevenly. We were unable to fit both the motor and the latch in the same vertical line at the center of the guard. Therefore, the motor was mounted off-center as close as possible to the latch, and a 3D printed linkage arm bolted through the guard with an arm extending to the nut. While this method did tug up the

guard, it produced a significant moment, causing the guard to jam on the way up and over-compress the spring on the right side. This made it exceedingly difficult to retract without the springs, and impossible once they were added in. Unfortunately, we did not have enough time to solve this problem. However, it does illustrate the need for the retraction system to act on the center of the guard. Two simple possibilities to solve this issue exist: one, have a second motor and screw assembly on the other side of the latch to balance out the moment on the other side while also doubling the retraction force, and two, have the shaft mounted centrally with the motor still off-centered, and use a small timing belt to transfer the torque through the guard's center of mass.

SENSOR SYSTEM

The electronic and sensing system was designed to detect an approaching human and deploy a safety guard to cut off access to the mower blades. The initial requirements specified a detection range of 0.3 m in front of the guard with minimal latency. Electronics and sensors were required to minimize false positives caused by environmental challenges such as grass movement, debris, and mower-induced vibrations.

To overcome these challenges, a mmWave sensor was chosen as the detection sensor because it could detect human presence based on the Doppler effect and the micro-Doppler waves caused by humans in small shifts of the body. Alongside this, the IMU, an accelerometer and gyroscopic sensor, was integrated into the system to estimate the velocity of the ride-on lawnmower and detect vibrations induced by the fast-spinning blades. To process the data collected by the sensors, an ESP32 microcontroller was chosen based on cost, compatibility, and ease of use. These electronics were wired on a simple breadboard using separate power supplies for the different voltages required. Additional considerations include wiring, pin configuration,

and different communication protocols used, like I2C and UART. Additional electronic components included a motor driver for the stepper motor used in the retraction system and an electronic rotary latch used in the release system.

The mmWave sensor was placed on a sensor mount designed to assess different sensor angles relative to the forward direction. The sensor mount was placed at the center of the guard and behind the motor housing and designed with a height of 5" to clear the housing and allow detection at 0.3m in front of the guard. This mount was rigidly mounted to the top of the motor housing to help reduce vibration with #8 wood screws. The breadboard was placed directly behind the sensor mount and connected by a 5V power supply for the microcontroller and 24V for the stepper motor, wired by jumper wires. The prototype breadboard and sensor mount can be seen in *Appendix C, Figs. C-1 and C-2*. Initially, the guard was reset by resetting the code. Afterwards, a push button was selected to reset the guard, emulating manual control on a ride-on lawnmower panel.

After iterations to the release system and retraction system, a relay was introduced to the electronic system, a different communication protocol was used for the motor driver, and a limit switch was used to determine when the guard was reset and at the home position. Furthermore, a protoboard was selected to minimize wiring issues and make electronics look neat. The wiring for the protoboard can be seen in *Appendix C, Figs. C-3 and C-4*. For the newly introduced parts, challenges of placement arose, and thus, the sensor mount was changed to also be used as the electronic housing. The housing used standoffs to allow the electronics and wiring to have airflow and reduce temperature. The electronic housing was 3D printed, and press-fit thread inserts were placed into the standoffs. The electronics used a variety of bolts due to the different

hole diameters of the protoboard and relay, and bolts remaining from other systems. The Electronic housing drawing can be seen in *Appendix A, Fig. A-12*.

Furthermore, to simplify the wiring, a Bluetooth-based reset button replaced the push button that was initially decided on. The ESP32 microcontroller allows Bluetooth connection using BLE and an app called nRF Connect, allowing the guard to be reset from a phone. Also, to try and emulate a real mower system and further iterate the prototype, we used a 12V battery which is standard for all ride-on lawnmowers. For the different power inputs needed, a buck and boost converter were selected, turning 12V to 5V for the microcontroller and 12V to 24V for the stepper motor. The new solenoid latch used the 12V directly from the battery. Within the microcontroller, the 5V was converted to 3.3V, needed for the other electronic components, which include the IMU, mmWave Sensor, motor driver, relay, and limit switch. The buck and boost converter was mounted to the top of the blade housing by two wood screws in the slots that they came with. The 12V battery was then mounted to the top with simple 3D printed brackets that can be seen in *Appendix A, Fig. A-14* and crimped wires with alligator clips to connect to the battery terminals.

The final sensing system consists of a 24GHz mmWave radar sensor and an MPU-6050 IMU, both integrated with an ESP32 microcontroller. The radar sensor is configured to monitor a forward-facing detection zone with an effective range of 1.0m from the guard. During testing, the range was increased from the initial specification of 0.3m to 1.0m due to latency issues. The radar measures the range, velocity, and energy, where energy represents reflected radar signal strength, which generally increases with the size and reflectivity of the detected object. A valid detection occurs when a target is within the range, moving toward the mower, and has sufficient energy to exceed the designed detection thresholds of 1.45m from the sensor (1.0m from the

guard), a velocity threshold of 0.15m/s toward the mower, and an energy threshold of 50,000. A Kalman filter is applied to the data gathered by the IMU to estimate the velocity of the mower, which is then used to calculate the true velocity of the detected object. The IMU helps filter out false detections and noise caused by vibrations.

The ESP32 processes sensor data. When a valid detection occurs, the relay is triggered to allow the guard to deploy. The system then waits for the Bluetooth button to reset. Once pressed, the system uses the mmWave sensor to determine if the area is clear. If the area is clear, the stepper motor retracts the guard until the limit switch is pressed. The stepper motor then reverses, allowing the lead screw to disengage, and the system returns to the idle state. The different system states and code can be seen in *Appendix D*.

All electronics are powered from a single 12V battery using buck and boost converters to supply the required voltages. The final system uses a soldered protoboard and a combined sensor mount and electronic housing to improve reliability, organization, and integration. A visual overview of the system can be seen in *Appendix C, Fig. C-5*.

TESTING

During our testing phase, we first analyzed the drop time of the guard. This is the time it takes for the guard to fall completely, starting the time as soon as it begins to fall. We measured this using the slow-motion feature of an iPhone camera, and after five trials, we found the average drop time to be $46\text{ms} \pm 7.07\text{ms}$. We then found the average total time. This was the time from detection to when the guard hit the ground. We measured it in the same way as the previous measurement, with the clock starting once a human enters detection range and ends when the guard hits the ground. We found our average total time after five trials to be $284\text{ms} \pm 7.07\text{ms}$. Subtracting the average drop time from the average total time, we found the average detection

time of the device to be $238\text{ms} \pm 10\text{ms}$. The calculation for the drop time, total time, and detection time can be seen below in *Figure 9*.

Video Time Trials					1 s	1000ms	Uncertainty							
Total Time														
	Start (sec)	Stop (sec)	Total (sec)	in ms			delta_t	0.005 sec						
Test 1	2.61	3.16	0.55	550	Video 2915		delta_q	0.007071 sec						
Test 2	2.45	2.53	0.08	80	Video 2911			7.071068 ms		Uncertainty for measured time				
Test 3	3	3.18	0.18	180	Video 2912		delta_Q	0.01 sec						
Test 4	3.05	3.61	0.56	560	Video 2913			10 ms		Uncertainty for calculated time				
Test 5	3.86	3.91	0.05	50	Video 2914									
		AVG	0.284	284										
Drop Time														
	Start	Stop	Total	in ms										
Test 1	3.11	3.16	0.05	50										
Test 2	2.48	2.53	0.05	50										
Test 3	3.15	3.18	0.03	30										
Test 4	3.56	3.61	0.05	50										
Test 5	3.86	3.91	0.05	50										
		AVG	0.046	46										
Detection Time														
	Start	Stop	Total	in ms										
Test 1	2.61	3.11	0.5	500										
Test 2	2.45	2.48	0.03	30										
Test 3	3	3.15	0.15	150										
Test 4	3.05	3.56	0.51	510										
Test 5	3.86	3.86	0	0										
		AVG	0.238	238										

Figure 9. Measurements and Calculations for Drop Time, Total Time, and Detection Time with Errors.

Next, we evaluated our device in different scenarios a lawn mower could face to see if the guard dropped when we wanted it to and did not drop when we did not want it to. The first two scenarios were scenarios in which we wanted the guard to drop. In the first scenario, we had a person stand still while the lawn mower moved toward them, and in the second scenario, a person moved toward the lawn mower while the lawn mower remained still. We ran five trials of each scenario, and the guard correctly dropped every single time. Next, we assessed scenarios in which we did not want the guard to drop as the user would theoretically not be in danger. The first one in which we kept the lawn mower still and had a person walk in front of it perpendicularly. Our device incorrectly released the guard in 5/5 trials. The final scenario we ran consisted of a person standing still and the lawn mower moving away from that person. The lawn mower correctly did not release in 5/5 trials. These results indicate that the lawn mower release

system is biased toward triggering incorrectly as opposed to being biased toward not triggering when it is supposed to. One of our specifications was that we wanted this device to be biased towards false positives, so although it is not working exactly how we expected it to, it is erring on the side of safety, which still meets the intent of our project.

SUMMARY AND CONCLUSIONS

Lawnmower-related injuries are the third leading cause of traumatic amputation among children. An estimated 9,000 pediatric injuries are reported annually. Riding lawnmowers are responsible for 76.1% of these injuries. Our group sought to address this problem by developing a mechanism to reduce the risk of injury. Our resulting prototype was “RALPH” — a radar-actuated blade guard that utilizes a millimeter-wave sensor array around the mower deck to detect the presence of humans or animals within a specified proximity and drops a protective guard. “RALPH” successfully integrated a variety of mechanical and electrical systems to create a functional lawnmowing safety device. We chose a 30”×5”×1/8” polycarbonate sheet to function as the guard to help with ease of assembly and for presentation purposes. The clear guard allows observers to see the components at work behind the guard during demonstrations. We would likely make this guard from aluminum in a real prototype. The guard’s vertical movement was guided by 7 mm linear rails. The rail carriages had 3D-printed spacers affixed to create more space for the retraction assembly. Stops were 3D-printed with PETG and affixed to the end of the rails to ensure the guard stays on the rails on uneven terrain.

The release system met its operational goals but has many areas for improvement. An ATOPLEE electronic rotary latch was mounted to the wooden base via a 3D-printed L-bracket. This bracket included gussets and a large base for structural integrity. The latch holds the guard up using a hook, secured to the polycarbonate guard via an aluminum U-channel. Springs were

utilized to ensure the strict safety timing requirements were met. We chose 81.5 mm (0.12 lbf/mm) compression springs that were positioned above the rail carriages. The rail carriages were redesigned to include a hole for a column to pass through, preventing spring buckling during retraction. We designed and 3D-printed a spring housing with gussets for structural integrity and wood-screwed it into the wooden base. The spring, which was compressed 50 mm, allowed for a closing time of around 40 ms.

The retraction system provided automatic retraction of the guard to reset the mechanism. The reset sequence is triggered using Bluetooth via a smartphone app. This emulates a physical button that would be placed on the mower control panel in a commercial product. It could also be made completely automatic once a threat has been determined to be “cleared;” however, we decided that this could lead to false resets. The current system ensures the user can verify the area is clear before resetting the device. The lifting motion is driven by a 24V NEMA 17 stepper motor, selected for its torque output and precise control. The motor is connected to a ball screw via a coupler and setscrews. This allows the nut to lift the guard by pushing on a 3D-printed connector. A limit switch on the latch signals the motor to stop once the guard has been relatched. Rubber siding was used to provide traction for the nut, preventing backsliding.

The sensing systems proved highly effective in distinguishing human presence. We used a 24GHz millimeter-wave radar sensor for its price and ability to distinguish between living things and inanimate objects. It does this by utilizing the Doppler effect to sense heartbeats. This radar was used along with an MPU-6050 IMU accelerometer and gyroscope to help gauge whether there was an imminent threat. It did this by sending raw data to an ESP32 microcontroller, which ran code to decide whether to drop the guard. To illustrate how this electronic system would interface with a real lawnmower, we powered the entire system with a

12V battery, the standard in all ride-on lawnmowers. Power converters were then used to step up or step down the voltage to power the different components.

The final prototype successfully integrated all these parts into a system capable of completely preventing user contact with lawnmower blades. Assessing the prototype against our initial design requirements, we met many of our key specifications. It is reusable and resettable. Its components are easily maintained. The prototype also proved to be highly cost-effective, costing less than \$600 to produce. This is well below 20% the cost of an average riding mower (\$3,940). This cost could be further lowered by leveraging bulk purchases and making certain parts in-house. This radar-based detection system reduces dependence on human behavior and awareness, creating a protection system for bystanders. Our team conducted multiple testing scenarios to gauge the robustness of our final design. Our current prototype appears to have a bias for false positives. The latch correctly released in all trials where a person moved towards the mower or the mower moved towards the person. It also successfully avoided dropping when the mower was moving away from a stationary person. The system still needs work; it incorrectly dropped in a situation where the test subject moved perpendicular to the device. The release system was highly effective: it took an average of $46 \text{ ms} \pm 7.07 \text{ ms}$ to drop the guard upon receiving a signal from the microcontroller.

Our prototype suffered two major challenges: sensor latency and mechanical binding during retraction. We aimed to have a total stopping time of 285 ms, i.e., from the moment something enters range to the moment the guard hits the ground. Our tests averaged the total stopping time to be $284 \text{ ms} \pm 7.07 \text{ ms}$. The sensor processing alone took an average of $238 \text{ ms} \pm 10 \text{ ms}$. We believe this is due to software and sensor latency — issues that could be fixed with code optimization and more efficient wiring. The mechanical systems were certainly not perfect

either. The retraction system experienced significant racking issues because of the off-center placement of the stepper motor and ball screw. We attempted to fix this by using a 3D-printed part to transmit the moment closer to the center. This racking made it exceedingly difficult to align the latch and often caused the guard to bind halfway during retraction. This binding may have been caused by the inability of the rails to manage off-axis loading or the spring pass-through columns clashing against the spacers. We were ultimately able to get the retraction system to work but had to remove the springs during the process.

Any future iterations must address reliability issues to ensure a smoother retraction cycle and a faster, more consistent detection time. The racking issue could be solved by using two motors for symmetrical load distribution or adding a third rail for alignment. We could also use a belt drive to transmit the motor's power to a centrally located shaft. Adding a latch on either side of the guard and moving the motor to the center could also prevent racking. Other future work includes incorporating a multi-sensor system to increase detection coverage on all sides and ensure better sensor reliability. We would also like to remake the guard to have 360° coverage. We would also add an automatic blade shutoff for added safety. To further this design, we would conduct more testing with future prototypes to better understand their strengths and weaknesses. The final major goal for this project is to assess our system on an actual mower in an outdoor setting to evaluate its viability in a real-world setting, and to contact user trials to test the effect on mowing performance and to see if there is a commercial market for this device.

APPENDIX A. Drawings

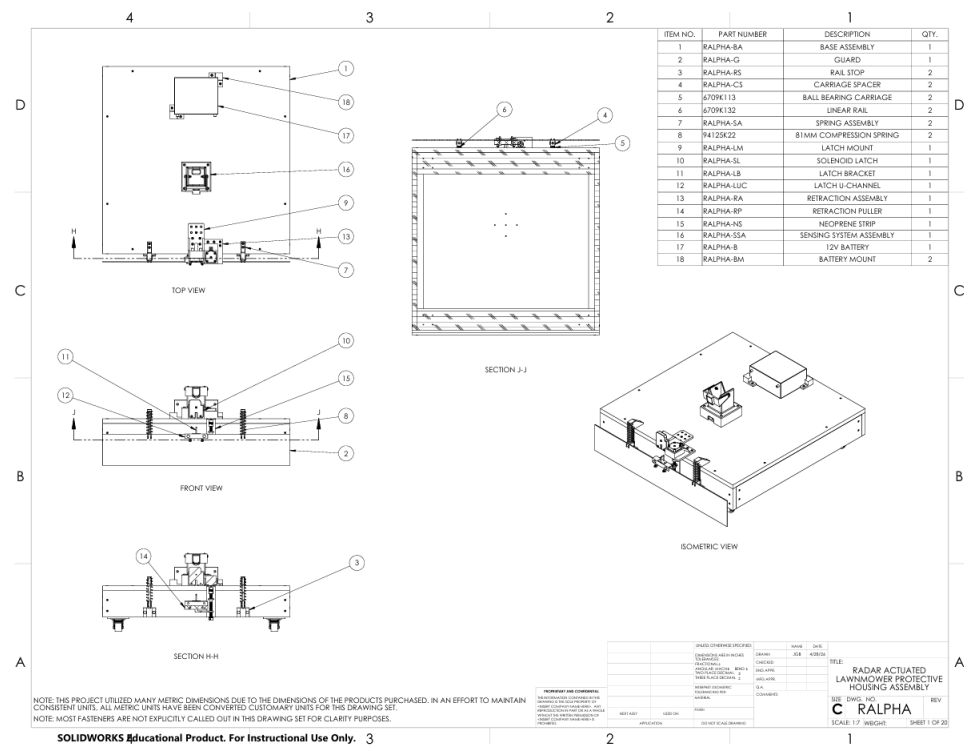


Figure A-1. Assembly Drawing

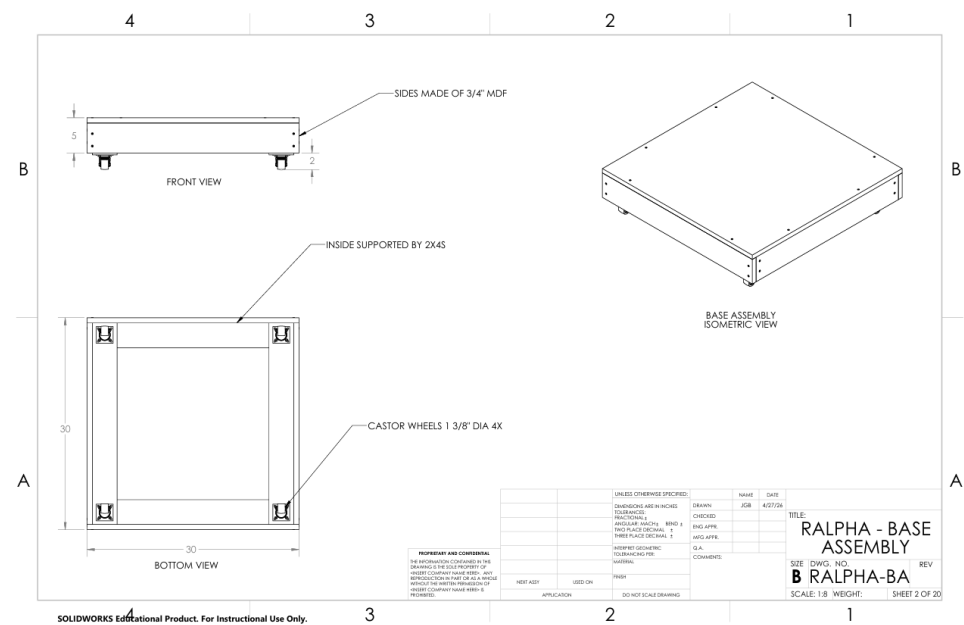


Figure A-2. Mockup Mower Deck Drawing

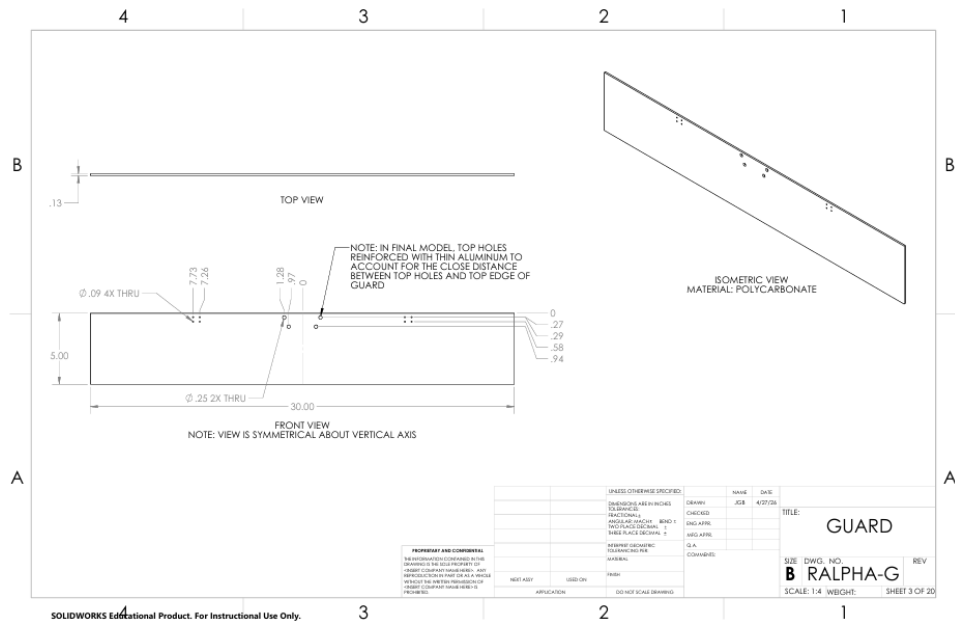


Figure A-3. Guard Drawing

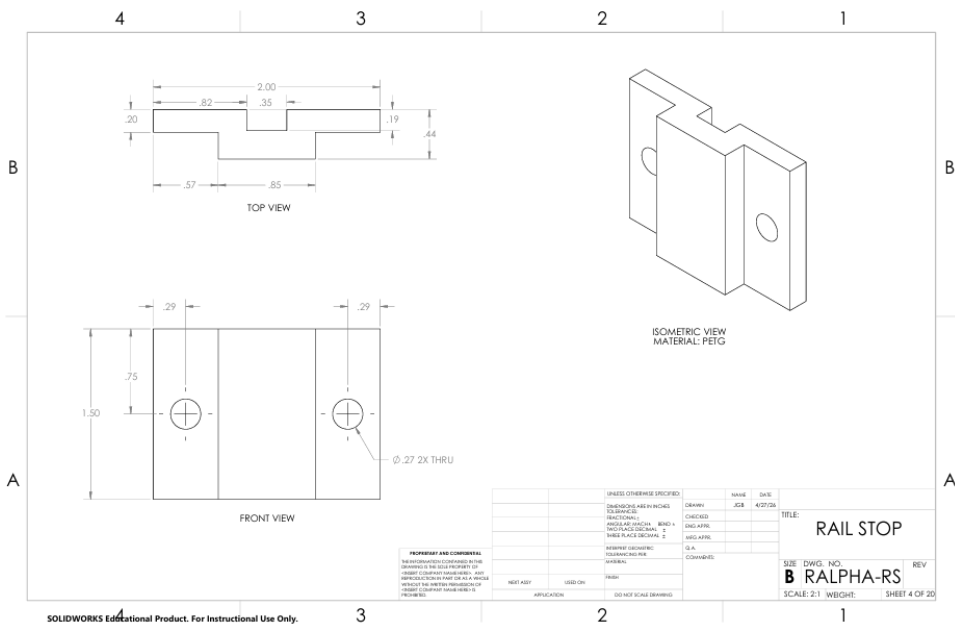


Figure A-4. Rail Stop Drawing

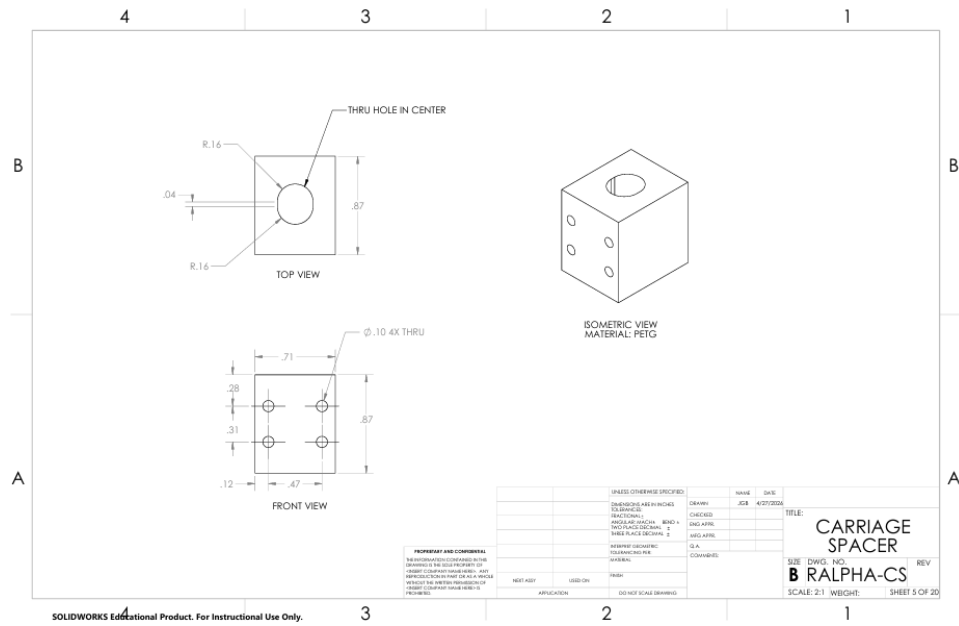


Figure A-5. Carriage Spacer Drawing

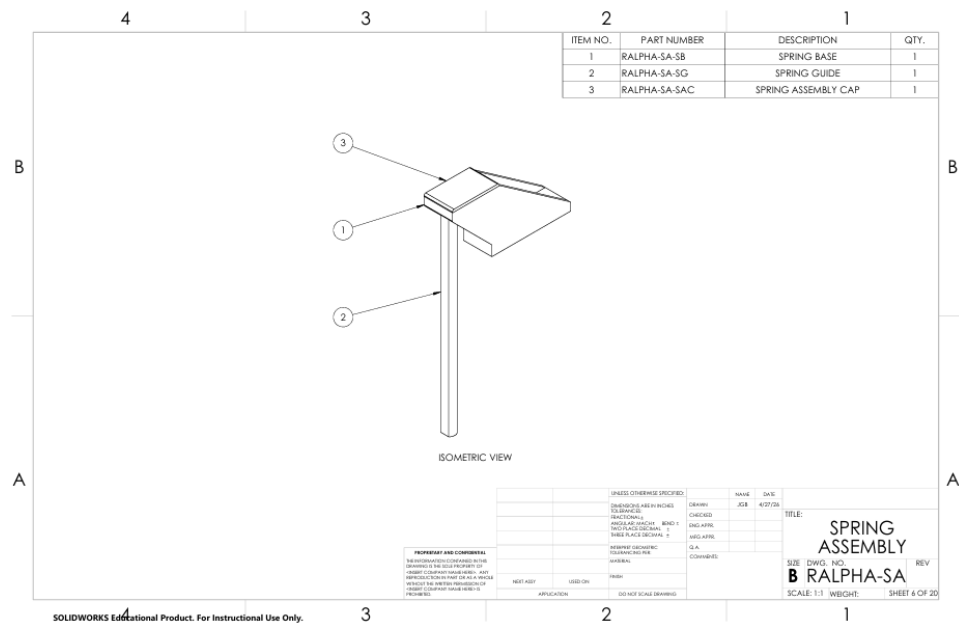


Figure A-6. Spring Assembly Drawing

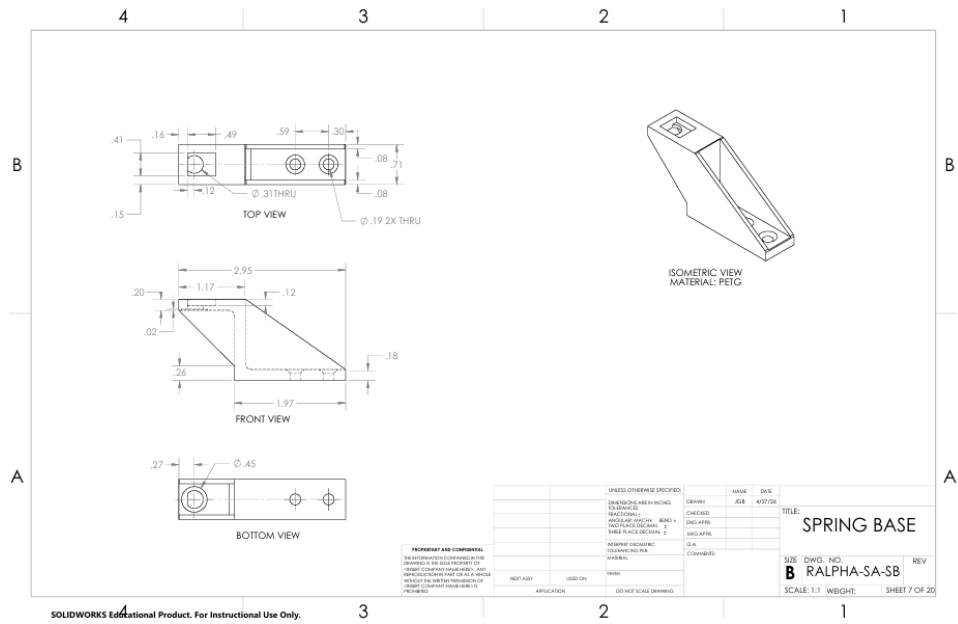


Figure A-6-1. Spring Base Drawing

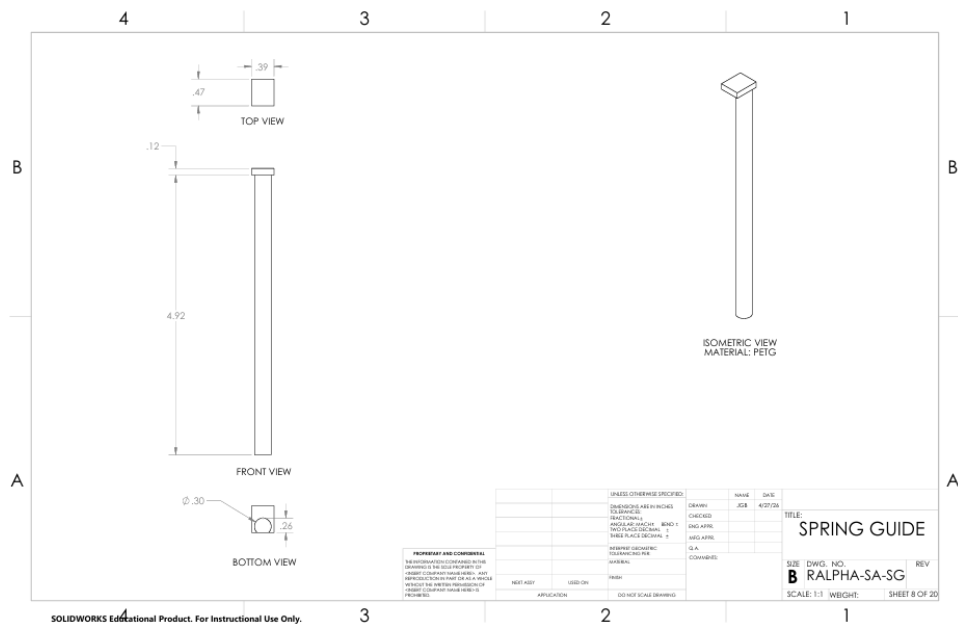


Figure A-6-2. Spring Guide Drawing

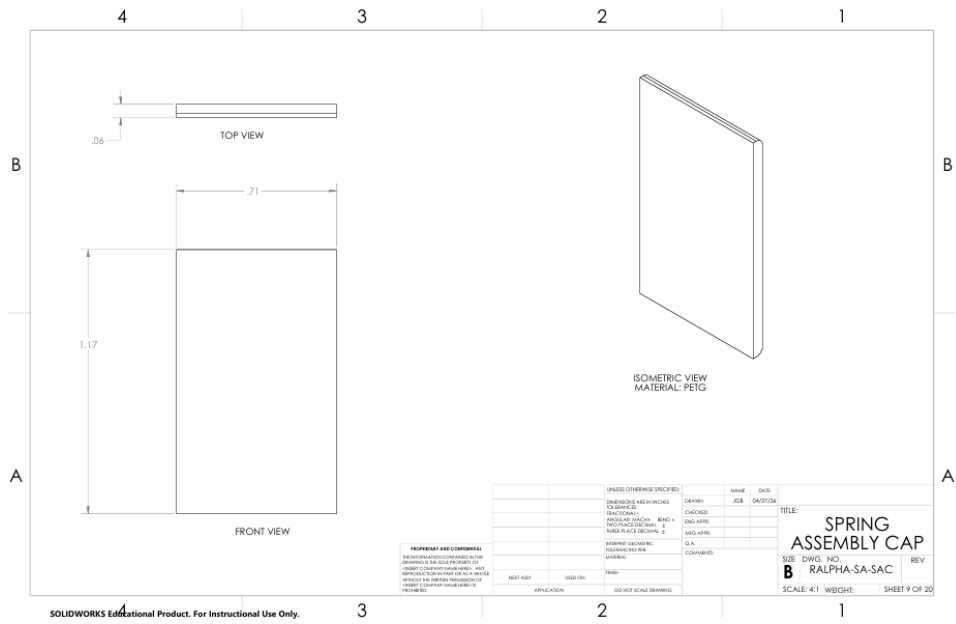


Figure A-6-3. Drawing of cap to be installed on the spring assembly

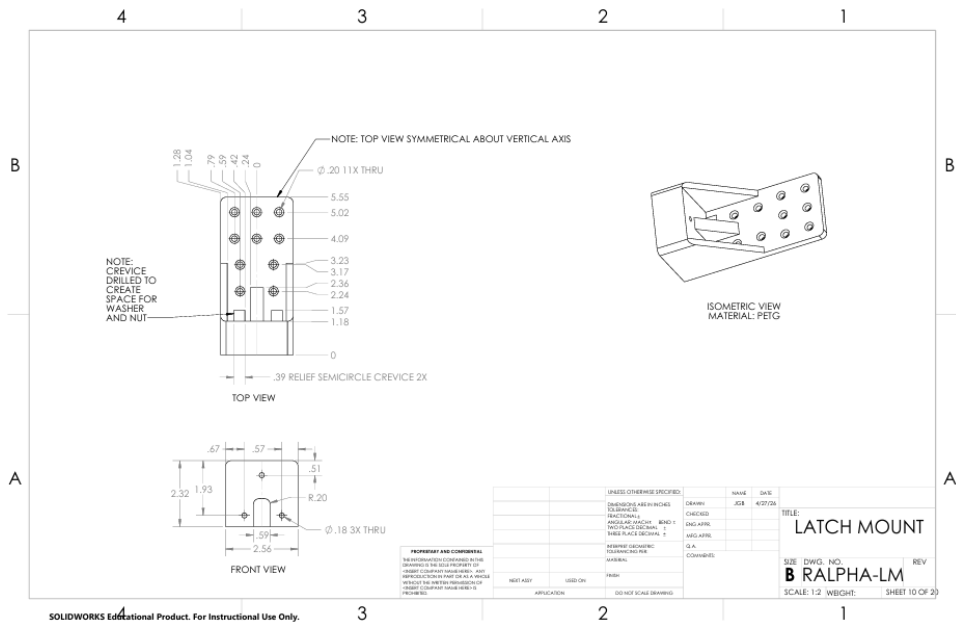


Figure A-7. Drawing of 3D printed mount to hold the electronic latch

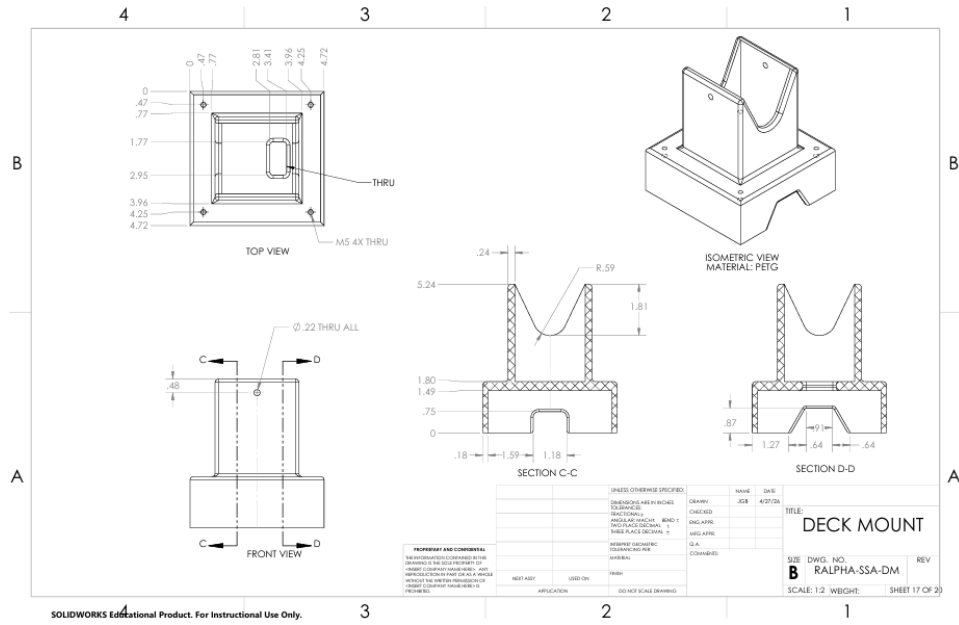


Figure A-12-1. Deck Mount Drawing

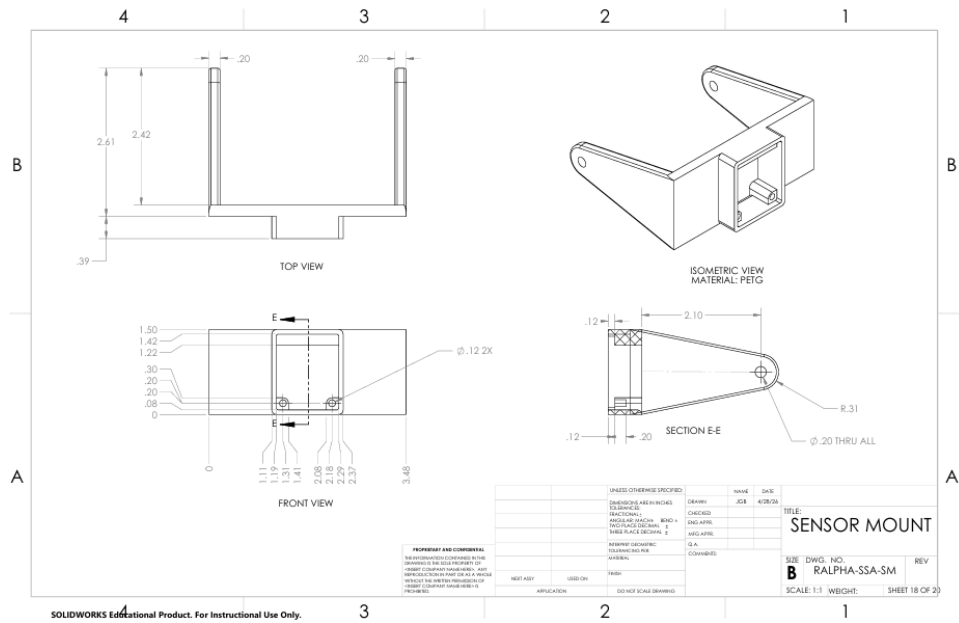


Figure A-12-2. Sensor Mount Drawing

APPENDIX B. Cost of Project

Figure B-1 shows the total cost of all purchases made, including those which did not make it into the final project. The total spent was \$935.54, which is under our budget.

Item	Number	Amount	Cost Per Unit	Description	Final Cost
TMC 2209 Motor Driver		1	\$ 8.95	Motor Driver	\$ 8.95
ESP32 Microcontroller	ESP-WROOM-32	1	\$ 9.99	To control all electronics	\$ 9.99
Ball Bearing Carriage	6709K113	2	\$ 66.00	Go up and down rails	\$ 132.00
130 mm rail	6709K132	2	\$ 68.89	Rails for guard to slide on	\$ 137.78
5/16-18 Nylon Lock nuts	90630A115	1	\$ 4.13	pack of 20	\$ 4.13
2-56 hex screws 1 1/8in long	91251A195	1	\$ 9.39	pack of 5	\$ 9.39
2-56 lock nut	90631A003	1	\$ 4.80	pack of 100	\$ 4.80
Aluminum U channel	9001K836	1	\$ 14.09	4 ft long	\$ 14.09
Electronic rotary release	5136N32	1	\$ 112.32	Initially selected latch	\$ 112.32
1/4-20 hex bolt	92865A551	1	\$ 15.37	per pack of 50	\$ 15.37
Thin-Profile Hex Nut	90695A025	1	\$ 6.44	For the Rails	\$ 6.44
18-8 Stainless Steel Socket Head Screw	91292A031	1	\$ 8.10	For the Rails	\$ 8.10
18-8 Stainless Steel General Purpose Washer	93475A195	1	\$ 1.78	For the Rails	\$ 1.78
Thrust Bearing		1	\$ 7.19	To support shaft	\$ 7.19
NEMA 17 Stepper motor	17HE15-1504S	1	\$ 4.72	To actuate screw	\$ 4.72
8mm leadscrew		1	\$ 23.31	To retract guard	\$ 23.31
mmWave - C4001	SEN0609	1	\$ 13.90	Our sensor	\$ 13.90
MPU-6050 IMU	B01DK83ZYQ	3	\$ 3.66	Measures acceleration and tilt	\$ 10.98
Breadboard Kit	EL-CK-003	1	\$ 15.99	For prototyping electronics	\$ 15.99
Molex Micro-Fit 3.0 male and female		1	\$ 7.98	for wiring the latch	\$ 7.98
Clear Impact-Resistant Polycarbonate	8574K263	1	\$ 24.72	Guard Material	\$ 24.72
30V 10A Power Supply		1	\$ 44.99	To test powering electronics	\$ 44.99
Caster Wheels 1 3/8 in dia	4778T51	4	\$ 18.86	for release 2 1/16in tall	\$ 75.44
1x1 Plywood		2	\$ 2.51	Assembling the base	\$ 5.02
MDF Board		1	\$ 49.98	To attach rails to	\$ 49.98
18-8 Stainless Steel Button Head Hex Drive Screw M5, 25mm Long	92095A216	2	\$ 4.16	for sensor base	\$ 8.32
18-8 Stainless Steel General Purpose Washer M5	93475A240	6	\$ 0.65	for sensor base	\$ 3.90
18-8 Stainless Steel Nylon-Insert Locknut M5	93625A200	2	\$ 4.46	for sensor base	\$ 8.92
18-8 Stainless Steel Button Head Hex Drive Screw M3, 8mm Long	92095A181	2	\$ 4.40	for sensor base	\$ 8.80
18-8 Stainless Steel Socket Head Screw	91292A034	1	\$ 3.78	For Rails (30mm)	\$ 3.78
Electronic solenoid latch		1	\$ 15.49	for releasing the guard	\$ 15.49
Protoboard kit		1	\$ 7.49	To solder electronics	\$ 7.49
3.3v relay module		1	\$ 9.95	To drive the latch	\$ 9.95
1/8" Neoprene	1374N15	1	\$ 6.68	Rubber w/ adhesive for stop	\$ 6.68
18-8 Stainless Steel Socket Head Screw	91292A034	1	\$ 3.78	For Carriages (30mm)	\$ 3.78
5mm to 8mm shaft coupling	B0DJ58YCYN	1	\$ 0.49	to join stepper motor and lead screw	\$ 0.49
4 pin JST connector kit		1	\$ 6.99	For connecting sensor	\$ 6.99
12v to 24V boost converter		1	\$ 14.99	Motor power supply	\$ 14.99
12v to 5v buck converter		1	\$ 7.99	ESP32 power supply	\$ 7.99
Limit switches		1	\$ 4.89	For detecting Lockout	\$ 4.89
4" Compression Springs		1	\$ 16.06	Initial Springs	\$ 16.06
81mm Compression Springs		3	\$ 3.11	Final Springs	\$ 9.33
EXTRA Electronic solenoid latch		1	\$ 15.49	Redundant Backup	\$ 15.49
EXTRA TMC 2209 Motor Driver		1	\$ 8.95	Redundant Backup	\$ 8.95
EXTRA ESP32 Microcontroller	ESP-WROOM-32	1	\$ 9.99	Redundant Backup	\$ 9.99
EXTRA mmWave - C4001	SEN0609	1	\$ 13.90	Redundant Backup	\$ 13.90

Figure B-1. Total Spent

Figure B-2 shows the itemized breakdown of expenses, only including those that actually made it to the final project, and calculating the unit cost for bulk items. The total cost of itemized expenses was \$593.82. It is worth noting that some items were attained for free (wood, woodscrews, and battery), and that many of these items would not be needed if applied to an

actual mower (such as the cost of the wood, wheels, and battery). Additionally, *Figure B-3* provides an estimate of the cost of 3D prints used in the final design (though it neglects the cost of discarded prints while prototyping).

Category	Item Name	# of Units	Cost per	Total Cost
Rail & Guard	Ball Bearing Carriage	2	\$ 66.00	\$ 132.00
Rail & Guard	130 mm rail	2	\$ 68.89	\$ 137.78
Rail & Guard	1x1 Plywood	2	\$ 2.51	\$ 5.02
Rail & Guard	2 x 4 Plywood	1	FREE	\$0.00
Rail & Guard	#8 Woodscrews	1	FREE	\$0.00
Rail & Guard	MDF board	1	\$ 49.98	\$ 49.98
Rail & Guard	Clear Impact-Resistant Polycarbonate	0.5	\$ 24.72	\$ 12.36
Rail & Guard	18-8 Stainless Steel General Purpose Washer	0.52	\$ 1.78	\$ 0.93
Rail & Guard	18-8 18mm Stainless Steel Socket Head Screw	0.2	\$ 8.10	\$ 1.62
Rail & Guard	Caster Wheels 1 3/8 in dia	4	\$ 18.86	\$ 75.44
Rail & Guard	18-8 30mm Stainless Steel Socket Head Screw	1	\$ 3.78	\$ 3.78
Rail & Guard	Thin-Profile Hex Nut	0.2	\$ 6.44	\$ 1.29
Release	aluminum U channel	0.125	\$ 14.09	\$ 1.76
Release	1/4-20 hex bolt	0.16	\$ 15.37	\$ 2.46
Release	Electronic solenoid latch	1	\$ 15.49	\$ 15.49
Release	81mm Compression Springs	2	\$ 3.11	\$ 6.22
Release	6061 1/16in Aluminum sheet	1	\$ 2.64	\$ 2.64
Release	2-56 hex screws 1 1/8in long	1	\$ 9.35	\$ 9.35
Retraction	TMC 2209 Motor Driver	1	\$ 8.95	\$ 8.95
Retraction	NEMA 17 Stepper motor	1	\$ 4.72	\$ 4.72
Retraction	8mm leadscrew	1	\$ 23.31	\$ 23.31
Retraction	5mm to 8mm shaft coupling	1	\$ 6.49	\$ 6.49
Retraction	Limit switches	0.1	\$ 4.89	\$ 0.49
Sensor and Electronics	ESP32 Microcontroller	1	\$ 8.95	\$ 8.95
Sensor and Electronics	mmWave - C4001	1	\$ 13.90	\$ 13.90
Sensor and Electronics	MPU-6050 IMU	1	\$ 3.66	\$ 3.66
Sensor and Electronics	Molex Micro-Fit 3.0 male and female	0.2	\$ 7.98	\$ 1.60
Sensor and Electronics	Protoboard kit	0.5	\$ 7.49	\$ 3.75
Sensor and Electronics	3.3v relay module	1	\$ 9.95	\$ 9.95
Sensor and Electronics	1/8" Neoprene	0.5	\$ 6.68	\$ 3.34
Sensor and Electronics	4 pin JST connector kit	0.1	\$ 6.99	\$ 0.70
Sensor and Electronics	12v to 24V boost converter	1	\$ 14.99	\$ 14.99
Sensor and Electronics	12v to 5v buck converter	1	\$ 7.99	\$ 7.99
Sensor and Electronics	12V Battery	1	FREE	\$0.00
Sensor and Electronics	18-8 Stainless Steel Button Head Hex Drive Screw M5, 25mm Long	1	\$ 4.16	\$ 4.16
Sensor and Electronics	18-8 Stainless Steel General Purpose Washer M5	1	\$ 0.65	\$ 0.65
Sensor and Electronics	18-8 Stainless Steel Nylon-Insert Locknut M5	1	\$ 4.46	\$ 4.46
Sensor and Electronics	18-8 Stainless Steel Button Head Hex Drive Screw M3, 8mm Long	1	\$ 4.40	\$ 4.40
Other	3D Prints	1	\$ 9.26	\$ 9.26

Figure B-2. Itemized Final Budget

Name	Volume (mm ³)	Cost *
Spring Holders	25903.32	\$ 0.28
Spring Support Rod	10889.22	\$ 0.12
Spacers	13996.38	\$ 0.15
Motor to Guard Connector	27659.07	\$ 0.30
Battery Holder	51127.64	\$ 0.56
Motor Mount	77667.01	\$ 0.85
Latch Mount	175997.07	\$ 1.94
Stoppers	29496.7	\$ 0.32
Sensor Housing	428668.42	\$ 4.72
Total:		\$ 9.26
* based on 50% infill, ABS cost of \$20/kg, density of 1100kg/m ³		

Figure B-3. Estimated Cost of 3D Prints in Final Design

APPENDIX C. Electronics and Sensing System

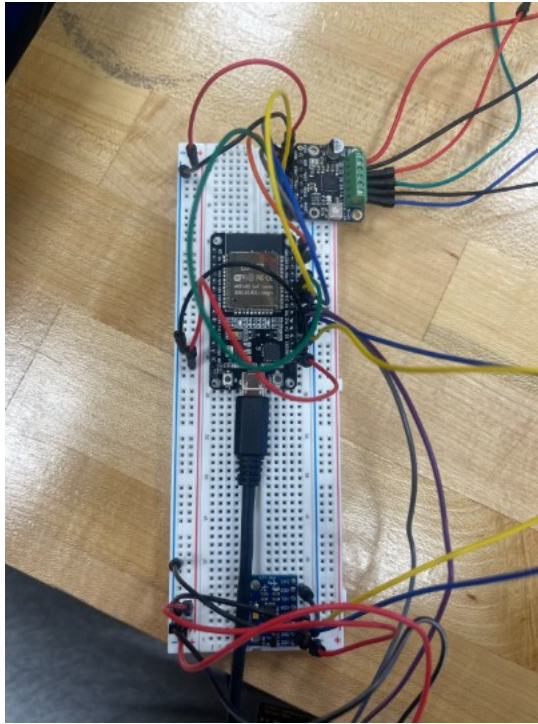


Figure C-1. Breadboard Wiring

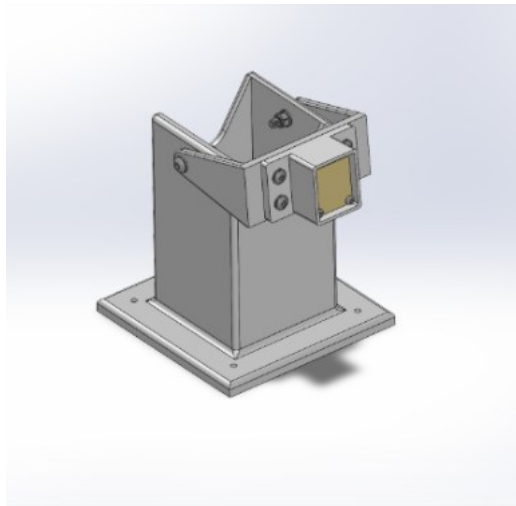


Figure C-2. Initial Sensor Mount

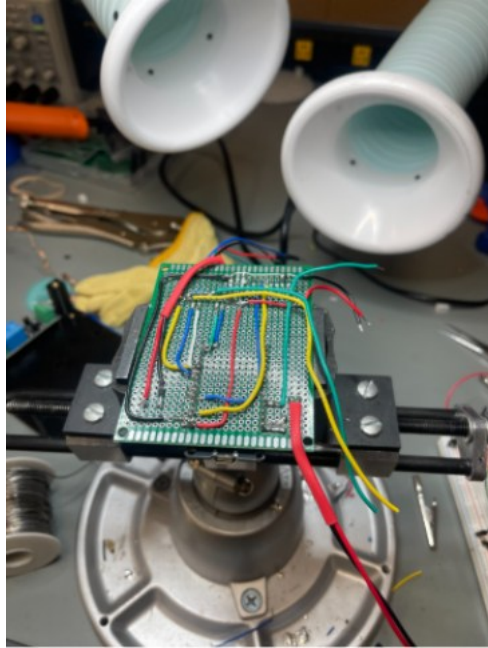


Figure C-3. Protoboard Bottom Side Wiring

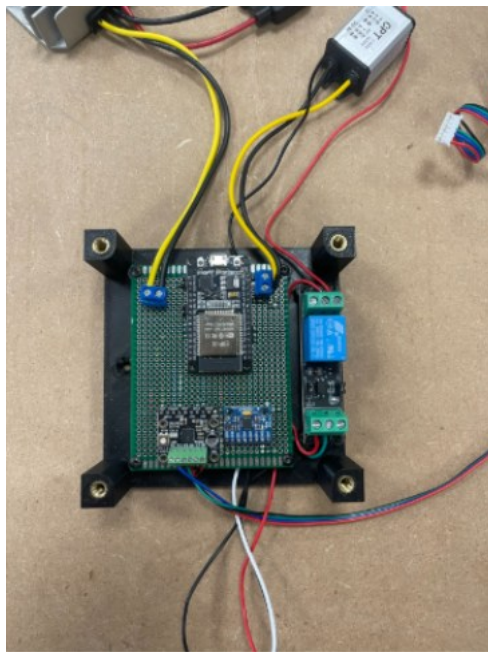


Figure C-4. Protoboard Top Side Wiring

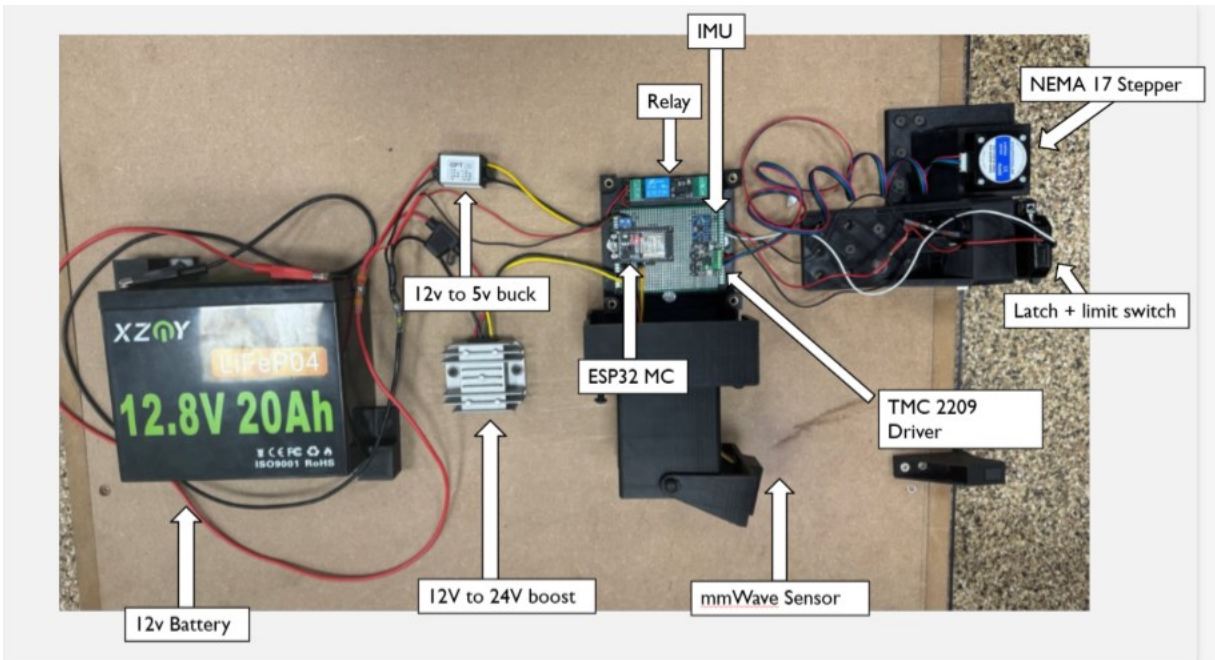


Figure C-5. Electronics and Sensing System Overview

APPENDIX D. Main Code

```
#include "Config.h"
#include "ApproachDetector.h"
#include "RelayLatch.h"
#include "StepperGuard.h"
#include "LimitSwitch.h"

#include <BLEDevice.h>
#include <BLEServer.h>
#include <BLEUtils.h>

ApproachDetector detector;
RelayLatch latch(RELAY_PIN, true, RELAY_PULSE_MS);
StepperGuard guard(Serial1);
LimitSwitch homeSwitch(LIMIT_SWITCH_PIN, true);

enum SystemState {
    STATE_IDLE,
    STATE_LATCH_FIRING,
    STATE_ARMED_FOR_BLE_RESET,
    STATE_RETRACTING,
    STATE_REVERSING,
    STATE_WAIT_CLEAR
};

SystemState systemState = STATE_IDLE;
unsigned long stateStartMs = 0;
unsigned long clearStartMs = 0;
unsigned long reverseStartMs = 0;

// BLE settings
#define BLE_DEVICE_NAME      "RALPH_Control"
#define SERVICE_UUID         "12345678-1234-1234-1234-123456789abc"
#define CHARACTERISTIC_UUID  "abcd1234-5678-1234-5678-abcdef123456"

BLECharacteristic* pCommandCharacteristic = nullptr;
bool bleClientConnected = false;
bool resetCommandRequested = false;

// Serial command input
String serialInput = "";

void setState(SystemState newState) {
    systemState = newState;
```

```

        stateStartMs = millis();
    }

    void fullSystemReset() {
        Serial.println("=== SERIAL RESET TRIGGERED ===");

        // Stop moving
        guard.hardStop();

        // Reset detector
        detector.reset();

        // Reset timers
        clearStartMs = 0;
        reverseStartMs = 0;
        resetCommandRequested = false;

        // Reset indicators
        digitalWrite(STATUS_LED_PIN, LOW);

        // Return to state 0
        setState(STATE_IDLE);

        Serial.println("System reset -> STATE_IDLE");
    }

    class MyServerCallbacks : public BLEServerCallbacks {
        void onConnect(BLEServer* pServer) {
            bleClientConnected = true;
            Serial.println("BLE client connected");
        }

        void onDisconnect(BLEServer* pServer) {
            bleClientConnected = false;
            Serial.println("BLE client disconnected");
            BLEDevice::startAdvertising();
            Serial.println("BLE advertising restarted");
        }
    };

    class CommandCallbacks : public BLECharacteristicCallbacks {
        void onWrite(BLECharacteristic* pCharacteristic) {
            String value = pCharacteristic->getValue().c_str();
            value.trim();

```

```

        if (value.length() > 0) {
            Serial.print("BLE received: ");
            Serial.println(value);
        }

        if (value == "RESET") {
            resetCommandRequested = true;
        }
    }
};

void setupBLE() {
    BLEDevice::init(BLE_DEVICE_NAME);

    BLEServer* pServer = BLEDevice::createServer();
    pServer->setCallbacks(new MyServerCallbacks());

    BLEService* pService = pServer->createService(SERVICE_UUID);

    pCommandCharacteristic = pService->createCharacteristic(
        CHARACTERISTIC_UUID,
        BLECharacteristic::PROPERTY_WRITE
    );

    pCommandCharacteristic->setCallbacks(new CommandCallbacks());

    pService->start();
    BLEDevice::startAdvertising();

    Serial.print("BLE ready. Device name: ");
    Serial.println(BLE_DEVICE_NAME);
}

void handleBLECommands() {
    if (!resetCommandRequested) {
        return;
    }

    resetCommandRequested = false;
    Serial.println("Phone requested RESET");

    if (systemState != STATE_ARMED_FOR_BLE_RESET) {
        Serial.println("RESET ignored: system not armed");
    }
}

```

```

        return;
    }

    detector.update();

    if (detector.isHumanApproaching()) {
        Serial.println("RESET blocked: human still approaching");
        digitalWrite(STATUS_LED_PIN, HIGH);
        return;
    }

    Serial.println("RESET accepted -> retracting guard");
    digitalWrite(STATUS_LED_PIN, LOW);
    guard.moveHome();
    setState(STATE_RETRACTING);
}

void handleSerialCommands() {
    while (Serial.available()) {
        char c = Serial.read();

        if (c == '\n') {
            serialInput.trim();

            if (serialInput.length() > 0) {
                Serial.print("Serial command: ");
                Serial.println(serialInput);
            }

            if (serialInput.equalsIgnoreCase("reset")) {
                fullSystemReset();
            }

            serialInput = "";
        }
        else if (c != '\r') {
            serialInput += c;
        }
    }
}

void setup() {
    Serial.begin(115200);

```

```

pinMode (STATUS_LED_PIN, OUTPUT);
digitalWrite (STATUS_LED_PIN, LOW);

detector.begin();
detector.reset();

latch.begin();
guard.begin();
homeSwitch.begin();

setupBLE();

setState (STATE_IDLE);

Serial.println("RALPH system initialized.");
Serial.println("Type 'reset' in Serial Monitor to return to
STATE_IDLE.");
}

void loop() {
    latch.update();
    guard.update();
    homeSwitch.update();

    handleSerialCommands();
    handleBLECommands();

    switch (systemState) {
        case STATE_IDLE:
            detector.update();

            digitalWrite (STATUS_LED_PIN,
                detector.isHumanApproaching() ? HIGH : LOW);

            if (detector.isHumanApproaching()) {
                Serial.println("Human detected -> releasing latch");
                latch.trigger();
                digitalWrite (STATUS_LED_PIN, HIGH);

                detector.reset();
                setState (STATE_LATCH_FIRING);
            }
            break;
    }
}

```

```

    case STATE_LATCH_FIRING:
        if (!latch.isActive()) {
            Serial.println("Latch pulse complete -> waiting for
RESET");

            detector.reset();
            digitalWrite(STATUS_LED_PIN, LOW);
            setState(STATE_ARMED_FOR_BLE_RESET);
        }
        break;

    case STATE_ARMED_FOR_BLE_RESET:
        detector.update();
        digitalWrite(STATUS_LED_PIN,
            detector.isHumanApproaching() ? HIGH : LOW);
        break;

    case STATE_RETRACTING:
        if (homeSwitch.isPressed()) {
            Serial.println("Switch pressed -> reverse NOW");
            guard.moveAway();
            reverseStartMs = millis();
            setState(STATE_REVERSING);
        }
        break;

    case STATE_REVERSING:
        if (millis() - reverseStartMs >= 2000) {
            Serial.println("Reverse complete -> STOP");
            guard.hardStop();

            detector.reset();
            clearStartMs = 0;
            digitalWrite(STATUS_LED_PIN, LOW);

            setState(STATE_WAIT_CLEAR);
        }
        break;

    case STATE_WAIT_CLEAR:
        detector.update();

        digitalWrite(STATUS_LED_PIN,
            detector.isHumanApproaching() ? HIGH : LOW);

```

```

        if (!detector.isHumanApproaching()) {
            if (clearStartMs == 0) {
                clearStartMs = millis();
            }

            if (millis() - clearStartMs >= 1000) {
                Serial.println("Area clear -> re-arming");
                detector.reset();
                digitalWrite(STATUS_LED_PIN, LOW);
                setState(STATE_IDLE);
            }
        } else {
            clearStartMs = 0;
        }
        break;
    }

    static unsigned long lastPrint = 0;
    if (millis() - lastPrint >= 250) {
        lastPrint = millis();

        Serial.print("State: ");
        Serial.print(systemState);

        Serial.print(" | Targets: ");
        Serial.print(detector.getTargetCount());

        Serial.print(" | Range(m): ");
        Serial.print(detector.getRangeM(), 3);

        Serial.print(" | mmWave Vel(m/s): ");
        Serial.print(detector.getRadarSpeedMps(), 3);

        Serial.print(" | IMU Vel(m/s): ");
        Serial.print(detector.getLawnmowerVelocityMps(), 3);

        Serial.print(" | True Vel(m/s): ");
        Serial.print(detector.getTrueVelocityMps(), 3);

        Serial.print(" | Energy: ");
        Serial.print(detector.getEnergy());

        Serial.print(" | Stationary: ");
        Serial.print(detector.isStationaryNow() ? "YES" : "NO");
    }

```

```
Serial.print(" | HumanApproaching: ");
Serial.print(detector.isHumanApproaching() ? "YES" : "NO");

Serial.print(" | Relay: ");
Serial.print(latch.isActive() ? "ON" : "OFF");
Serial.print(" | Latched: ");
Serial.print(homeSwitch.isPressed() ? "YES" : "NO");

Serial.print(" | Limit: ");
Serial.print(homeSwitch.isPressed() ? "PRESSED" : "OPEN");

Serial.print(" | BLE: ");
Serial.println(bleClientConnected ? "CONNECTED" : "WAITING");
}
}
```


APPENDIX E. References

Bernini, F. (2009). *Lawn-mower* (US Patent) [Review of *Lawn-mower*].

<https://patents.google.com/patent/US20080282658A1/en>

Jennissen, C. A., Krupp, T. D., Vakkalanka, J. P., & Hoogerwerf, P. J. (2023). Pediatric lawn mower-related injuries and contributing factors for bystander injuries. *Injury Epidemiology*, 10(Suppl 1), 51. <https://doi.org/10.1186/s40621-023-00468-z>

Safekids Worldwide (2020, November). *Bicycling Injuries Among Children Ages 0–19 in 2020* [Review of *Bicycling Injuries Among Children Ages 0–19 in 2020*].
https://www.safekids.org/sites/default/files/documents/fast_facts_-_2022_-_bicycling_injuries.pdf

United States Plastic Corp. “What Guidelines Must Be Followed When Drilling Holes in Acrylic Sheets for Mounting? | U.S. Plastic Corp.” *Usplastic.com*, 2023,
www.usplastic.com/knowledgebase/article.aspx?contentkey=882.